

CSC458 PA2

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1 Questions

1. Larger buffers mean the queuing time for a packet can increase as the larger queue gets full without dropping packets, increasing the fetch times. Additionally, since RTT is measured on ACK, having smaller buffers means more dropped packets which don't affect RTT measurements as much.

2.
$$\text{max queue time} = \frac{\text{bits in queue}}{\text{draining rate}} = \frac{1000 \text{ txqueuelen} \cdot 1500 \text{ byte packet size} \cdot \frac{8 \text{ bits}}{\text{byte}}}{100 \cdot 1,000,000 \text{ bps}} = 0.12 \text{ seconds} = 120 \text{ ms}$$

3.
$$\text{RTT} = \min RTT + \frac{\text{Queue size}}{\text{Bottleneck bandwidth}}$$

CWND and RTT oscillations are strongly correlated. In Q100, plots are in phase, as the spikes in RTT are explained by the changing CWND sizes. As higher CWND allows more packets in flight, the queue gets more full and each packet must wait longer in the queue and its measured RTT increases.

Having smaller buffers can affect TCP performance by reducing latency. It is better for webpage fetching since having a lower latency is preferred for user experience.

Having larger buffers allow for higher throughput and utilization over long-lived flows. For TCP connections this may be desired, but for short-lived webpage fetches higher throughput and utilization from increasing CWND cannot be achieved.

4. The problem with larger buffers and simple drop tail algo is that the sender only knows if there is congestion at the router when the buffer is full. If the buffer is too large, this congestion signal is too late, allowing a long queue to build up and cause more delay.

Random Early Detection (RED): introduce packet drop probability proportional to the number of packets in the queue, along with min and max thresholds of the average queue length that define when the probability is used.

Explicit Congestion Notification (ECN): instead of dropping packet like in RED, the router forwards the packet with ECN flag set, and the receiver keeps the ECN flag in the ACK to signal the sender.

The experiment would similarly use long-lived TCP connection and webpage fetches workload through a control drop tail queuing mechanism with queue sizes 20 and 100. Then we repeat the experiments with RED and ECN by enabling them in the link's TCIntf config.

2 Theoretical Analysis

1. One cycle is linear increase from W_{min} to W_{max} , then it drops back down to W_{min} . Since each group of window sized packets that get ACK'd increases the CWND by 1, the total number of packets sent in one cycle is the summation of a linear term as follows:

$$\begin{aligned}
\sum_{i=W_{max}/2}^{W_{max}} i &= \sum_{i=0}^{W_{max}/2} (W_{max}/2 + i) \\
&= \sum_{i=0}^{W_{max}/2} (W_{max}/2) + \sum_{i=0}^{W_{max}/2} i \\
&= \frac{W_{max}}{2} + \sum_{i=1}^{W_{max}/2} (W_{max}/2) + \sum_{i=0}^{W_{max}/2} i \\
&= \frac{W_{max}}{2} + \left(\frac{W_{max}}{2}\right)^2 + \frac{W_{max}/2 \cdot (W_{max}/2 + 1)}{2} \\
&= \frac{W_{max}}{2} + \left(\frac{W_{max}^2}{4}\right) + \frac{W_{max}^2/4 + W_{max}/2}{2} \\
&= \frac{W_{max}}{2} + \left(\frac{W_{max}^2}{4}\right) + \frac{(W_{max}^2)}{8} + \frac{W_{max}}{4} \\
&= \frac{3W_{max}^2}{8} + \frac{3W_{max}}{4} \\
&\in O\left(\frac{3W_{max}^2}{8}\right)
\end{aligned}$$

2. First, observe that $\bar{W} = \frac{W_{min} + W_{max}}{2} = \frac{3W_{max}}{4}$ or $W_{max} = \frac{4\bar{W}}{3}$

Since loss is observed after $N_{packets}$ are sent,

$$\begin{aligned}
p &= \frac{1}{N_{packets}} \\
\Rightarrow p &\approx \frac{8}{3W_{max}^2} \\
\Rightarrow W_{max} &\approx \sqrt{\frac{8}{3p}} \\
\Rightarrow \frac{4\bar{W}}{3} &\approx \sqrt{\frac{8}{3p}} \\
\bar{W} &\approx \sqrt{\frac{3}{2}} \frac{1}{\sqrt{p}}
\end{aligned}$$

Given $T = \frac{\text{transfer size}}{\text{transfer time}}$ and the above two observations,

$$\begin{aligned}
T &= \frac{\bar{W} \cdot MSS}{RTT} \\
\Rightarrow T &\approx .8 \frac{\sqrt{\frac{3}{2}} \frac{1}{\sqrt{p}} \cdot MSS}{RTT} \\
\Rightarrow T &\approx \frac{\sqrt{\frac{3}{2}} \cdot MSS}{RTT \sqrt{p}} \\
\Rightarrow K &\approx \sqrt{\frac{3}{2}}
\end{aligned}$$

3. RTT can be expressed as

$$\begin{aligned}
RTT &= T_{prop} + T_{queue} \\
RTT &= RTT_0 + Q \cdot \tau
\end{aligned}$$

C can be expressed in units of bits per second as C_{bps} .

Then, τ can be expressed as $\tau = \frac{MSS \cdot 8}{C_{bps}}$

And the max queuing delay is when $Q = B$,

Together, we get RTT as

$$\begin{aligned}
RTT &= RTT_0 + B \cdot \tau \\
RTT &= RTT_0 + B \cdot \frac{MSS \cdot 8}{C_{bps}}
\end{aligned}$$

4. As seen from Q3, RTT is directly proportional to B. So increasing B increases short-transfer latency. But in long term flows the CWND increases so that $T \approx C$ so it remains high.